



Time Series Analysis

Seminar 1: Stochastic processes and stationarity



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **GDP**: Gross Domestic Product
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **MA**: Moving Average
- ▣ **PACF**: Partial AutoCorrelation Function

Today's question and route

- **Question:** is a series stationary, is it white noise, and how do the ACF and the Ljung–Box test answer these questions?
 - ▶ this seminar comes **before** Lecture 1: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: autocovariances of moving averages and of a random walk, a sample ACF and a Ljung–Box statistic by hand
 - ▶ Part B: BET, S&P 500 and EUR/RON returns; Romanian GDP and inflation, each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	mean, autocovariances and ACF of MA(1) and MA(2) processes	Solved, Proposed	A1
A3, A4	a random walk; a trend-stationary series and its first difference	Solved, Proposed	A3
A5, A6	sample ACF, bands, Box–Pierce and Ljung–Box by hand	Solved, Proposed	A5
B1, B2	prices and returns: BET; S&P 500 and EUR/RON	Solved, Proposed	B1
B3, B4	transformations: Romanian real GDP; Romanian HICP inflation	Solved, Proposed	B3
C1, C2	has inflation become more persistent? what is wrong in an AI answer?	Proposed	B1, B4

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
BET, S&P 500	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, not adjusted	1995–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	1996–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped; the last day is 18 September 2026
- Growth and inflation rates in %: $100 \Delta \ln Y_t$ (one period), $100 \Delta_4 \ln Y_t$ and $100 \Delta_{12} \ln P_t$ (one year)
- In the notebook: `b1_returns('bet')`, `sample_acf(x)`, `ljung_box(x, 10)`; no account or key is needed

Prerequisites for Today (1/15): processes and moments

- **Stochastic process** $\{X_t\}$: one random variable for each date t
 - ▶ a time series $x_1, \dots, x_T$ is one observed path of the process
 - ▶ $E[\cdot]$: the expected value, an average over all possible paths at a fixed date

- **Mean and autocovariance**

$$\mu_t = E[X_t], \quad \gamma(t, s) = \text{Cov}(X_t, X_s) = E[(X_t - \mu_t)(X_s - \mu_s)]$$

- ▶ $\gamma(t, s) > 0$: when X_t is above its mean, X_s tends to be above its mean too; $\gamma(t, t) = \text{Var}(X_t)$
 - ▶ correlation: $\text{Corr}(X_t, X_s) = \gamma(t, s) / \sqrt{\text{Var}(X_t) \text{Var}(X_s)}$, between -1 and 1
- **Rules for covariances**, with constants a, b, c
 - ▶ $\text{Cov}(aX + bY, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)$; $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$
 - ▶ $\text{Var}(\sum_i a_i X_i) = \sum_i \sum_j a_i a_j \text{Cov}(X_i, X_j)$; for uncorrelated X_i : $\sum_i a_i^2 \text{Var}(X_i)$

Prerequisites for Today (2/15): stationarity

- **Weakly stationary:** finite variance, constant mean, autocovariance that depends only on the lag

$$E[X_t^2] < \infty, \quad E[X_t] = \mu, \quad \text{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{for every } t$$

- ▶ h : the lag, the distance in periods between two dates; $\gamma(0) = \text{Var}(X_t)$
- ▶ a shift in time does not change the mean, the variance or the covariances

- **ACF** (autocorrelation function): the autocovariance in units of the variance

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}, \quad \rho(0) = 1, \quad |\rho(h)| \leq 1, \quad \rho(-h) = \rho(h)$$

- **Strictly stationary:** every joint distribution is unchanged by a shift in time
 - ▶ $(X_{t_1}, \dots, X_{t_k})$ and $(X_{t_1+h}, \dots, X_{t_k+h})$ have the same distribution, for every k , dates $t_1, \dots, t_k$ and shift h
 - ▶ **i.i.d.** (independent and identically distributed): the X_t are independent and all have the same distribution

Prerequisites for Today (3/15): white noise and moving averages

- **White noise** $\varepsilon_t \sim \text{WN}(0, \sigma^2)$: unpredictable shocks

$$E[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_s) = 0 \quad (t \neq s)$$

- ▶ σ^2 : the variance of a shock
- ▶ weak (only uncorrelated), i.i.d. (independent), Gaussian (i.i.d. Normal)

- **MA(q)** (moving average): a weighted sum of the current and the last q shocks

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

- ▶ μ : the mean; $\theta_1, \dots, \theta_q$: the weights of the past shocks; q : the order

- **Autocovariances of an MA(q)**: only the shocks common to X_t and X_{t+h} count

$$\gamma(h) = \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h} \quad (0 \leq h \leq q), \quad \gamma(h) = 0 \quad (h > q)$$

- ▶ with $\theta_0 = 1$; X_t and X_{t+h} share the shocks $\varepsilon_{t+h-q}, \dots, \varepsilon_t$
- ▶ beyond lag q no shock is shared: the ACF cuts off after lag q

Prerequisites for Today (4/15): random walk, AR(1) and trends

- **Random walk:** each value is the previous one plus a shock

$$X_t = X_{t-1} + \varepsilon_t, \quad X_0 = 0 \quad \implies \quad X_t = \varepsilon_1 + \dots + \varepsilon_t$$

- ▶ with drift: $X_t = c + X_{t-1} + \varepsilon_t$; c : the drift, the average change per period
- ▶ every shock stays in all later values

- **AR(1)** (autoregressive): $X_t = \phi X_{t-1} + \varepsilon_t$, with $|\phi| < 1$

- ▶ ϕ : the share of today's value that passes to tomorrow; $\rho(h) = \phi^h$, a geometric decay
- ▶ $\phi = 1$ gives the random walk

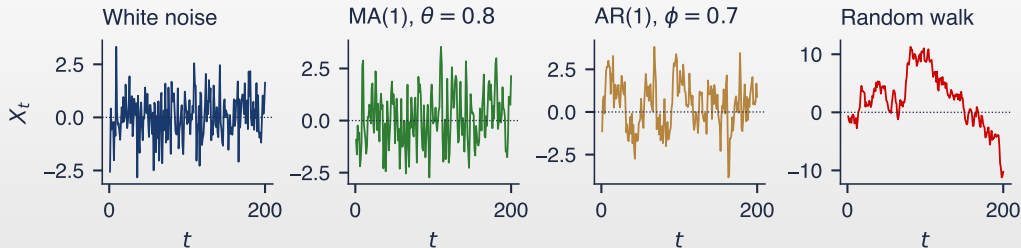
- **Trend-stationary:** $Y_t = a + bt + u_t$, with u_t stationary

- ▶ $a + bt$: a deterministic line; b : its slope, the average change per period

- **Differences**

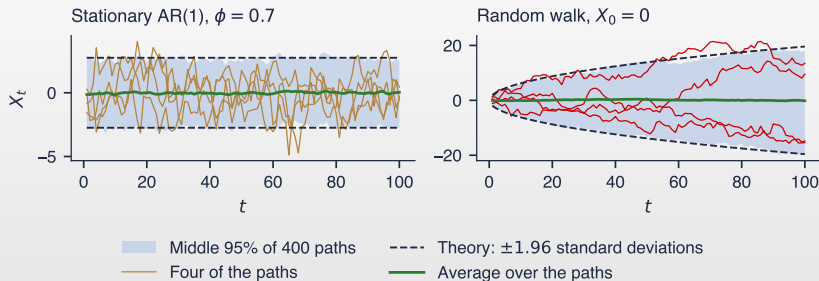
- ▶ $\Delta X_t = X_t - X_{t-1}$; seasonal: $\Delta_s X_t = X_t - X_{t-s}$, s : the number of periods in a year

Prerequisites for Today (5/15): one path of four processes



- ▣ Simulated with the same kind of shocks, $\varepsilon_t \sim N(0, 1)$, $T = 200$
- ▣ From left to right, more and more memory
 - ▶ white noise: no pattern; MA(1): neighbours are similar, the memory lasts one period
 - ▶ AR(1) with $\phi = 0.7$: longer waves, but always back around 0
 - ▶ random walk: no level to return to; the path wanders far from 0

Prerequisites for Today (6/15): many paths at once



- 400 simulated paths of each process; blue band: the middle 95% of the paths at each date t
- The mean and the variance at a fixed t are computed across the paths
 - AR(1): the band has the same width at every t , $\pm 1.96 \sqrt{1/(1 - \phi^2)} = \pm 2.74$: stationary
 - random walk: the band widens like $\pm 1.96 \sqrt{t}$; the variance grows with t : not stationary

Prerequisites for Today (7/15): the sample ACF and its band

- **Sample autocovariance and sample ACF** (divisor T)

$$\hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-h} (x_t - \bar{x})(x_{t+h} - \bar{x}), \quad \hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

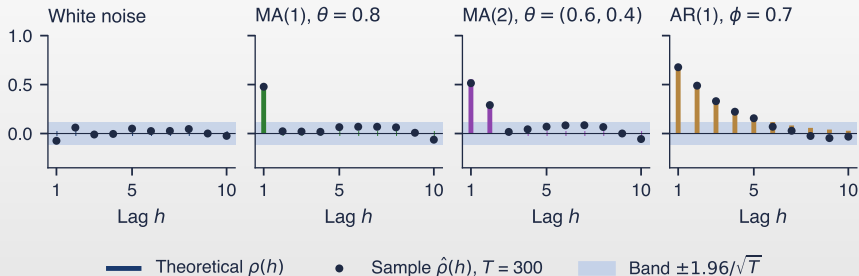
- ▶ $\bar{x} = \frac{1}{T} \sum_t x_t$: the sample mean; the hat marks a value estimated from the data
- ▶ the sample ACF estimates $\rho(h)$ only if the series is stationary

- **Under i.i.d. data:** $\hat{\rho}(h) \approx N(0, 1/T)$, with standard error $1/\sqrt{T}$; the **95% band** is

$$\pm \frac{1.96}{\sqrt{T}}$$

- ▶ 1.96: the 97.5% quantile of the standard Normal distribution
- ▶ a bar outside the band: $\rho(h) \neq 0$ at 5%, for that lag alone
- ▶ with 20 lags, about 1 bar crosses the band by chance

Prerequisites for Today (8/15): theoretical and sample ACF

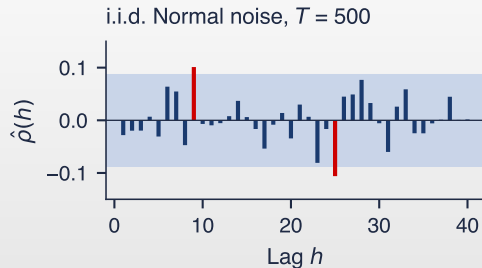


□ Bars: the theoretical $\rho(h)$; dots: $\hat{\rho}(h)$ from one simulated path, $T = 300$

□ Typical patterns

- ▶ white noise: no bar; MA(1): one bar, $\rho(1) = \theta/(1 + \theta^2) = 0.49$; MA(2): two bars, then nothing
- ▶ AR(1): $\rho(h) = 0.7^h$, a geometric decay
- ▶ random walk or trend: a very slow, almost linear decay; seasonal data: peaks at lags $s, 2s, \dots$

Prerequisites for Today (9/15): the band on i.i.d. noise



- Band $\pm 1.96/\sqrt{T} = \pm 0.088$
- Inside the band
- Outside, by chance

- One simulated i.i.d. Normal series, $T = 500$: band $\pm 1.96/\sqrt{500} = \pm 0.088$
- 2 of the 40 bars (lags 9 and 25) cross the band
 - expected: 5% of 40 lags is 2 lags
 - a single bar just outside the band is weak evidence; the portmanteau tests look at all lags together
- Significant is not large**
 - $\hat{\rho}(1)^2$: about the share of the variance of x_t that a linear forecast from x_{t-1} explains
 - with $T = 5000$, $\hat{\rho}(1) = 0.03$ is outside the band, yet explains only 0.09% of the variance

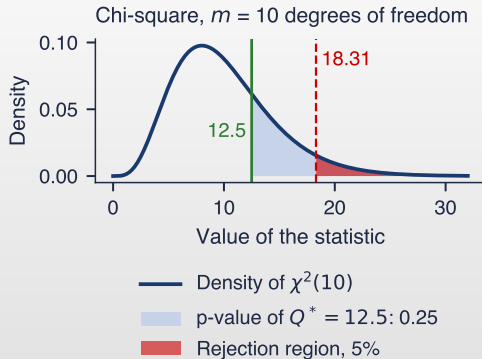
Prerequisites for Today (10/15): portmanteau tests

- H_0 : $\rho(1) = \dots = \rho(m) = 0$, the series is white noise up to lag m
 - ▶ m : the number of lags tested; H_1 : at least one $\rho(h) \neq 0$
- **Box–Pierce** and **Ljung–Box**: sums of squared sample autocorrelations

$$Q(m) = T \sum_{h=1}^m \hat{\rho}(h)^2, \quad Q^*(m) = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}(h)^2}{T-h}$$

- ▶ squares: positive and negative autocorrelations both raise the statistic
- ▶ the factor $(T+2)/(T-h) > 1$ corrects the small-sample behaviour of Q
- ▶ under H_0 both are approximately $\chi^2(m)$, the chi-square distribution with m degrees of freedom
- **Decision**: reject H_0 at 5% if the statistic exceeds $\chi_{0.95}^2(m)$, the 95% quantile of $\chi^2(m)$
 - ▶ 5% critical values: $\chi_{0.95}^2(2) = 5.99$, $\chi_{0.95}^2(3) = 7.81$, $\chi_{0.95}^2(8) = 15.51$, $\chi_{0.95}^2(10) = 18.31$, $\chi_{0.95}^2(12) = 21.03$

Prerequisites for Today (11/15): critical value and p-value

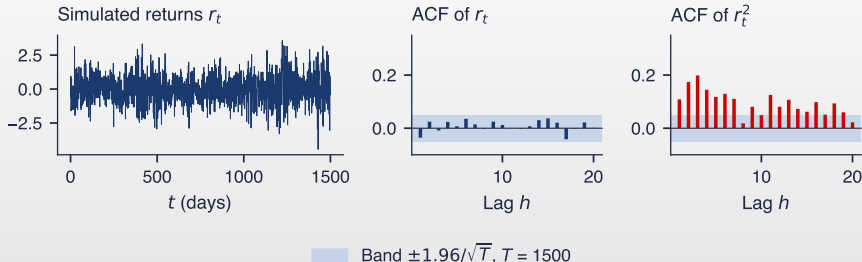


- **p-value:** the probability, under H_0 , of a statistic at least as large as the observed one

$$p = P(\chi^2(m) \geq Q_{\text{obs}}^*)$$

- ▶ Q_{obs}^* : the value computed from the data; p : the area to the right of it
- ▶ reject H_0 at 5% when $p < 0.05$, the same as $Q_{\text{obs}}^* > \chi_{0.95}^2(m)$
- **Chart:** $m = 10$
 - ▶ critical value 18.31: 5% of the area lies to its right (red)
 - ▶ $Q_{\text{obs}}^* = 12.5$ gives $p = 0.25$ (blue area): do not reject white noise
 - ▶ the mean of $\chi^2(m)$ is m : values near m are what H_0 expects

Prerequisites for Today (12/15): volatility clustering: returns and their squares



- **Volatility clustering:** large moves, of either sign, come in groups
 - ▶ simulated daily returns r_t whose variance depends on yesterday's squared return, $T = 1500$
- Two ACFs of the same series
 - ▶ r_t : $\hat{\rho}(1) = -0.04$, all bars near the band: the direction of a move is not predictable
 - ▶ r_t^2 : $\hat{\rho}(1) = 0.11$, bars outside the band up to lag 20: the size of a move is predictable
 - ▶ so r_t is uncorrelated, but not independent

Prerequisites for Today (13/15): transformations

- **Logarithm** of a positive series Y_t : $\ln Y_t$
 - ▶ stabilises a variance that grows with the level; a constant growth rate becomes a straight line
- **Growth rates** in % from the log

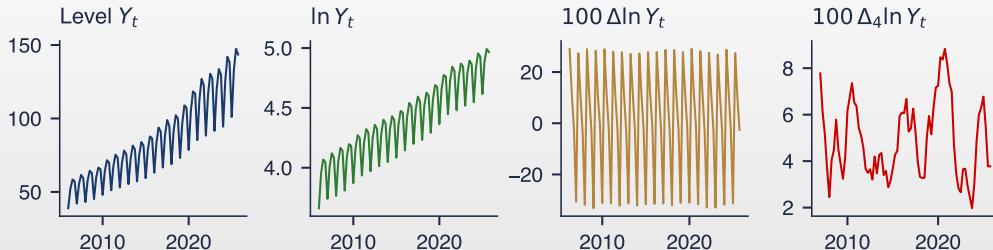
$$100 \Delta \ln Y_t = 100 (\ln Y_t - \ln Y_{t-1}), \quad 100 \Delta_4 \ln Y_t = 100 (\ln Y_t - \ln Y_{t-4})$$

- ▶ Δ : growth over one period; Δ_4 (Δ_{12}): growth over a year for quarterly (monthly) data
 - ▶ differences remove trends and random walks; Δ_4 and Δ_{12} also remove the season
- **Box-Cox**: a family of power transformations

$$Y_t^{(\lambda)} = \frac{Y_t^\lambda - 1}{\lambda} \quad (\lambda \neq 0), \quad Y_t^{(0)} = \ln Y_t$$

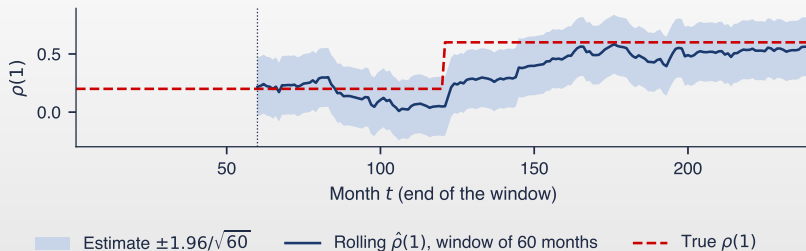
- ▶ λ : the power, chosen so that the variance becomes stable; $\lambda = 1$: no change in shape

Prerequisites for Today (14/15): transformations of a growing seasonal series



- ▣ A simulated quarterly series, 2006–2025, growing and seasonal; four views of the same data
- ▣ What each transformation does
 - ▶ level: the seasonal swing grows with the level; $\ln Y_t$: the swing has a constant size
 - ▶ $100 \Delta \ln Y_t$: the trend is gone, the season dominates (standard deviation 21.9%)
 - ▶ $100 \Delta_4 \ln Y_t$: trend and season are gone; mean 4.9%, standard deviation 1.7%

Prerequisites for Today (15/15): rolling-window estimates



- **Rolling window** of length w : the statistic is recomputed on the last w observations at each date t
 - ▶ $\hat{\rho}_t(1)$: the sample $\hat{\rho}(1)$ of $x_{t-w+1}, \dots, x_t$; it shows whether the persistence changes over time
- **Chart**: a simulated AR(1) whose ϕ jumps from 0.2 to 0.6 at month 120; $w = 60$
 - ▶ the estimate follows the change with a delay of up to w months
 - ▶ standard error about $1/\sqrt{w} = 0.13$; shaded: the estimate ± 1.96 standard errors

A1: an MA(1) process [Solved]

Task

- ▣ $X_t = \varepsilon_t + 0.5 \varepsilon_{t-1}$, with $\varepsilon_t \sim \text{WN}(0, 2)$.
- ▣ 1. Compute the mean $E[X_t]$.
- ▣ 2. Compute $\gamma(0)$, $\gamma(1)$ and $\gamma(2)$.
- ▣ 3. Compute $\rho(1)$ and $\rho(2)$.
- ▣ 4. Say whether the process is weakly stationary, and why.
- ▣ Report: five numbers and one sentence.

Solution

- ▣ 1. $E[X_t] = 0 + 0.5 \cdot 0 = 0$
- ▣ 2. $\gamma(0) = \text{Var}(\varepsilon_t) + 0.25 \text{Var}(\varepsilon_{t-1}) = 2 + 0.5 = 2.50$
- ▣ $\gamma(1) = \text{Cov}(\varepsilon_t + 0.5\varepsilon_{t-1}, \varepsilon_{t+1} + 0.5\varepsilon_t) = 0.5 \cdot 2 = 1.00$;
 $\gamma(2) = 0$ (no common shock)
- ▣ 3. $\rho(1) = 1.00/2.50 = 0.40$; $\rho(2) = 0$
- ▣ 4. Yes: the mean and all $\gamma(h)$ are finite and do not depend on t .

A2: an MA(2) process [Proposed]

Solution discussed in class

Task

- ▣ $X_t = 2 + \varepsilon_t + 0.4 \varepsilon_{t-1} - 0.3 \varepsilon_{t-2}$, with $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A1.
- ▣ 1. Compute the mean.
- ▣ 2. Compute $\gamma(0)$, $\gamma(1)$, $\gamma(2)$ and $\gamma(3)$.
- ▣ 3. Compute $\rho(1)$ and $\rho(2)$, and sketch the ACF up to lag 4.
- ▣ 4. Say how the constant 2 changes the autocovariances.
- ▣ Report: six numbers, the sketch and one sentence.

A3: a random walk [Solved]

Task

- ▣ $X_t = X_{t-1} + \varepsilon_t$, $X_0 = 0$, $\varepsilon_t \sim \text{WN}(0, 0.25)$.
- ▣ 1. Compute $E[X_{100}]$, $\text{Var}(X_{100})$ and $\text{Var}(X_{120})$.
- ▣ 2. Compute $\text{Cov}(X_{100}, X_{120})$ and $\text{Corr}(X_{100}, X_{120})$.
- ▣ 3. Say whether X_t and ΔX_t are stationary.
- ▣ 4. With a drift $c = 0.1$, compute $E[X_{100}]$.
- ▣ Report: six numbers and two sentences.

Solution

- ▣ 1. $X_t = \sum_{i=1}^t \varepsilon_i$: $E[X_{100}] = 0$;
 $\text{Var}(X_{100}) = 100 \cdot 0.25 = 25$; $\text{Var}(X_{120}) = 30$
- ▣ 2. The common shocks are $\varepsilon_1, \dots, \varepsilon_{100}$: $\text{Cov} = 25$;
 $\text{Corr} = 25 / \sqrt{25 \cdot 30} = \sqrt{100/120} = 0.913$
- ▣ 3. X_t : no, its variance grows with t ; $\Delta X_t = \varepsilon_t$: yes, it is white noise.
- ▣ 4. $E[X_{100}] = 100 \cdot 0.1 = 10$; the variance does not change.



A4: a trend-stationary series [Proposed]

Solution discussed in class

Task

- $Y_t = 5 + 0.2t + \varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A3.
- 1. Compute $E[Y_t]$ and $\text{Var}(Y_t)$, and say whether Y_t is stationary.
- 2. Write $D_t = \Delta Y_t$ and compute its mean, $\gamma(0)$, $\gamma(1)$ and $\rho(1)$.
- 3. Compare Y_t with a random walk with drift 0.2: what happens to a shock ε_{50} in each?
- Report: five numbers and two sentences.

A5: a sample ACF by hand [Solved]

Task

- ▣ Eight observations: $x = (4, 6, 5, 8, 7, 9, 6, 7)$.
- ▣ 1. Compute $\bar{x}$ and the deviations $x_t - \bar{x}$.
- ▣ 2. Compute $\hat{\gamma}(0)$, $\hat{\gamma}(1)$, $\hat{\gamma}(2)$ with divisor $T = 8$, then $\hat{\rho}(1)$ and $\hat{\rho}(2)$.
- ▣ 3. Compute the band $\pm 1.96/\sqrt{T}$ and compare.
- ▣ 4. Compute the Ljung–Box $Q^*(2)$ and decide at 5%.
- ▣ Report: seven numbers and one sentence.

Solution

- ▣ 1. $\bar{x} = 52/8 = 6.5$; deviations $(-2.5, -0.5, -1.5, 1.5, 0.5, 2.5, -0.5, 0.5)$
- ▣ 2. sum of squares 18: $\hat{\gamma}(0) = 2.250$; products at lag 1 sum to 0.25: $\hat{\gamma}(1) = 0.0312$; at lag 2 to 7.00: $\hat{\gamma}(2) = 0.875$
- ▣ $\hat{\rho}(1) = 0.014$, $\hat{\rho}(2) = 0.389$
- ▣ 3. Band ± 0.693 : both inside
- ▣ 4. $Q^*(2) = 8 \cdot 10 (\hat{\rho}(1)^2/7 + \hat{\rho}(2)^2/6) = 2.02 < 5.99$ ($p = 0.36$): do not reject white noise; with $T = 8$ the test has almost no power.

A6: Box–Pierce or Ljung–Box? [Proposed]

Solution discussed in class

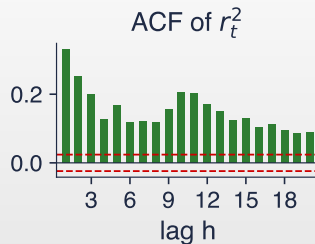
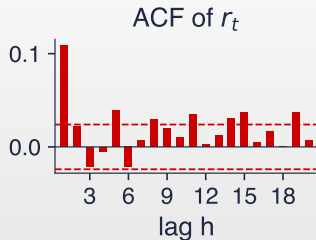
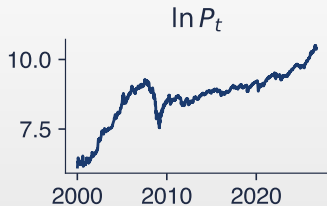
Task

- ▣ A series of $T = 120$ monthly returns has $\hat{\rho}(1) = 0.21$, $\hat{\rho}(2) = -0.08$, $\hat{\rho}(3) = 0.12$. Model: A5.
- ▣ 1. Compute the band and say which autocorrelations lie outside it.
- ▣ 2. Compute $Q(3)$ and $Q^*(3)$.
- ▣ 3. Compare both with $\chi^2_{0.95}(3) = 7.81$ and decide.
- ▣ 4. Explain why the two tests can disagree, and which one you trust here.
- ▣ Report: four numbers and two sentences.

B1: BET prices and returns [Solved]

- ▣ **Question:** is the BET log price stationary, and are BET daily returns white noise?
- ▣ **Inputs:** BET closes since 2000; $\ln P_t$ and $r_t = 100 \Delta \ln P_t$
- ▣ **Tasks**
 1. Compute the sample ACF of $\ln P_t$ at lags 1, 10 and 50.
 2. Draw the ACF of r_t up to lag 20 with the band $\pm 1.96/\sqrt{T}$, and count the bars outside it.
 3. Compute Ljung–Box $Q^*(10)$ for r_t and for r_t^2 , with their p-values.
 4. Interpretation: can yesterday's return be used to forecast today's return?
- ▣ **Report:** three ACF values, the chart, two statistics with p-values and two sentences
- ▣ **Notebook:** section B1

B1: solution (1/2) [Solved]



----- $\pm 1.96/\sqrt{T}$ ■ sample ACF

B1: solution (2/2) [Solved]

- $\ln P_t$: $\hat{\rho}(1) = 0.999$, $\hat{\rho}(10) = 0.991$, $\hat{\rho}(50) = 0.953$: almost no decay, a random-walk-like level, not stationary
- r_t ($T = 6\,670$, band ± 0.024): 7 of 20 bars outside, the largest $\hat{\rho}(1) = 0.109$; $Q^*(10) = 108.0$, $p < 0.001$
- r_t^2 : $\hat{\rho}(1) = 0.33$; $Q^*(10) = 2440$, $p < 0.001$: much stronger dependence
- Interpretation: yesterday's return explains only $\hat{\rho}(1)^2 = 1.2\%$ of today's variance, too little for a trading rule after costs; the size of yesterday's move does predict today's risk

B2: the S&P 500 and EUR/RON [Proposed]

- ▣ **Question:** do the S&P 500 and the EUR/RON rate behave like the BET?
- ▣ **Inputs:** S&P 500 closes since 2000; EUR/RON (BNR reference rate) since July 2005; model: B1
- ▣ **Tasks**
 1. Repeat the three steps of B1 for both series.
 2. Compare the sign and the size of $\hat{\rho}(1)$ of the returns across BET, S&P 500 and EUR/RON.
 3. Compare $Q^*(10)$ of the squared returns across the three series.
 4. Interpretation: which of the three return series is closest to white noise?
- ▣ **Report:** one table (three series, six numbers each) and two sentences
- ▣ **Notebook:** section B2

B3: Romanian real GDP [Solved]

- ▣ **Question:** which transformation makes Romanian quarterly GDP closest to stationary?
- ▣ **Inputs:** real GDP, chain-linked volumes (2010), not seasonally adjusted, Eurostat, from 1995
- ▣ **Tasks**
 1. Compute $\ln Y_t$, $100 \Delta \ln Y_t$ and $100 \Delta_4 \ln Y_t$.
 2. Draw the three series and their ACF up to lag 12.
 3. Report $\hat{\rho}(1)$, $\hat{\rho}(4)$ and $Q^*(8)$ for each.
 4. Interpretation: why is $\hat{\rho}(4)$ of the quarterly growth rate close to 1?
- ▣ **Report:** a table of nine numbers, the chart and two sentences
- ▣ **Notebook:** section B3

B3: solution (1/2) [Solved]



B3: solution (2/2) [Solved]

	$\hat{\rho}(1)$	$\hat{\rho}(4)$	$Q^*(8)$
$\ln Y_t$	0.68	0.94	495.0
$100 \Delta \ln Y_t$	-0.24	0.96	329.6
$100 \Delta_4 \ln Y_t$	0.63	0.03	74.2

- $\ln Y_t$: trend and season, peaks at lags 4 and 8; $\Delta \ln Y_t$: the season dominates (standard deviation 27.5%); $\Delta_4 \ln Y_t$: mean 2.73%, the ACF dies out after 2–3 quarters
- Interpretation: a q/q rate of unadjusted data repeats the same seasonal swing every year, hence $\hat{\rho}(4) \approx 1$; the annual rate removes the season but reacts late to turning points and overlaps across quarters (positive $\hat{\rho}(1)$)

B4: Romanian inflation [Proposed]

- **Question:** is monthly Romanian inflation since 2005 white noise around its mean?
- **Inputs:** HICP, 2015 = 100, Eurostat, January 2005 to the last month available; model: B3
- **Tasks**
 1. Compute the monthly inflation $100 \Delta \ln P_t$ and the 12-month inflation $100 \Delta_{12} \ln P_t$.
 2. Draw the ACF of both up to lag 36 and report $\hat{\rho}(1)$, $\hat{\rho}(12)$ and $\hat{\rho}(24)$.
 3. Compute Ljung–Box $Q^*(12)$ for the monthly inflation.
 4. Interpretation: what does the ACF of the monthly inflation say about how fast an inflation shock fades?
- **Report:** six ACF values, one statistic with its p-value, the chart and two sentences
- **Notebook:** section B4

C1: has Romanian inflation become more persistent? [Proposed]

- ▣ **Question:** is monthly inflation more persistent after 2020 than in 2005–2019?
- ▣ **Inputs:** monthly HICP inflation of B4, split in January 2020; models: B1, B4
- ▣ **Tasks**
 1. Compute $\hat{\rho}(1)$ and $Q^*(12)$ in each subperiod, with the band of each.
 2. Draw $\hat{\rho}(1)$ on rolling windows of 60 months.
 3. List the events of 2020–2024 that could change the persistence.
 4. Interpretation: why is the difference of two sample autocorrelations hard to judge with 80 observations?
- ▣ **Report:** a table, the chart and a plan for a project
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to interpret the S&P 500 daily data since 2000 ($T = 6\,714$ returns). The answer:
- (a) The ACF of the log price at lag 1 is 0.999, so the log price is a stationary process with a very long memory.
- (b) The return autocorrelation at lag 1 is -0.099 , outside the band of 0.024, so a trader can forecast most of tomorrow's return.
- (c) Ljung-Box $Q(10)$ of the squared returns is 6133, so the returns themselves are autocorrelated.
- (d) A weakly stationary process is always strictly stationary, since its mean and variance are constant.
- (e) Differencing a series that is already stationary does no harm: white noise stays white noise.
- (f) A random walk has $\text{Var}(X(t)) = t * \sigma^2$, so it is not stationary.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ Autocovariances of linear processes follow from the covariance rules: only common shocks count
- ▣ A random walk has a growing variance; its difference is white noise; a trend-stationary series becomes an over-differenced MA(1)
- ▣ Look at the level, the changes and the squares: prices wander, returns are almost uncorrelated, squares are strongly correlated
- ▣ For macro data, choose the transformation first ($\log$, Δ , Δ_4 , Δ_{12}), then read the ACF
- ▣ An AI answer is a draft: check what is stationary, what “significant” means, and whether squares or levels were tested

After the seminar

- Lecture 1 develops each topic of today: stationarity, white noise and the random walk, Wold, ergodicity, ACF and PACF, the portmanteau tests, transformations
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation persistence in Romania and in the region (Hungary, Poland, Czechia)
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 1–2; [Hyndman and Athanasopoulos \(2021\)](#), Ch. 2–3; [Brockwell and Davis \(2016\)](#), Ch. 1
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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